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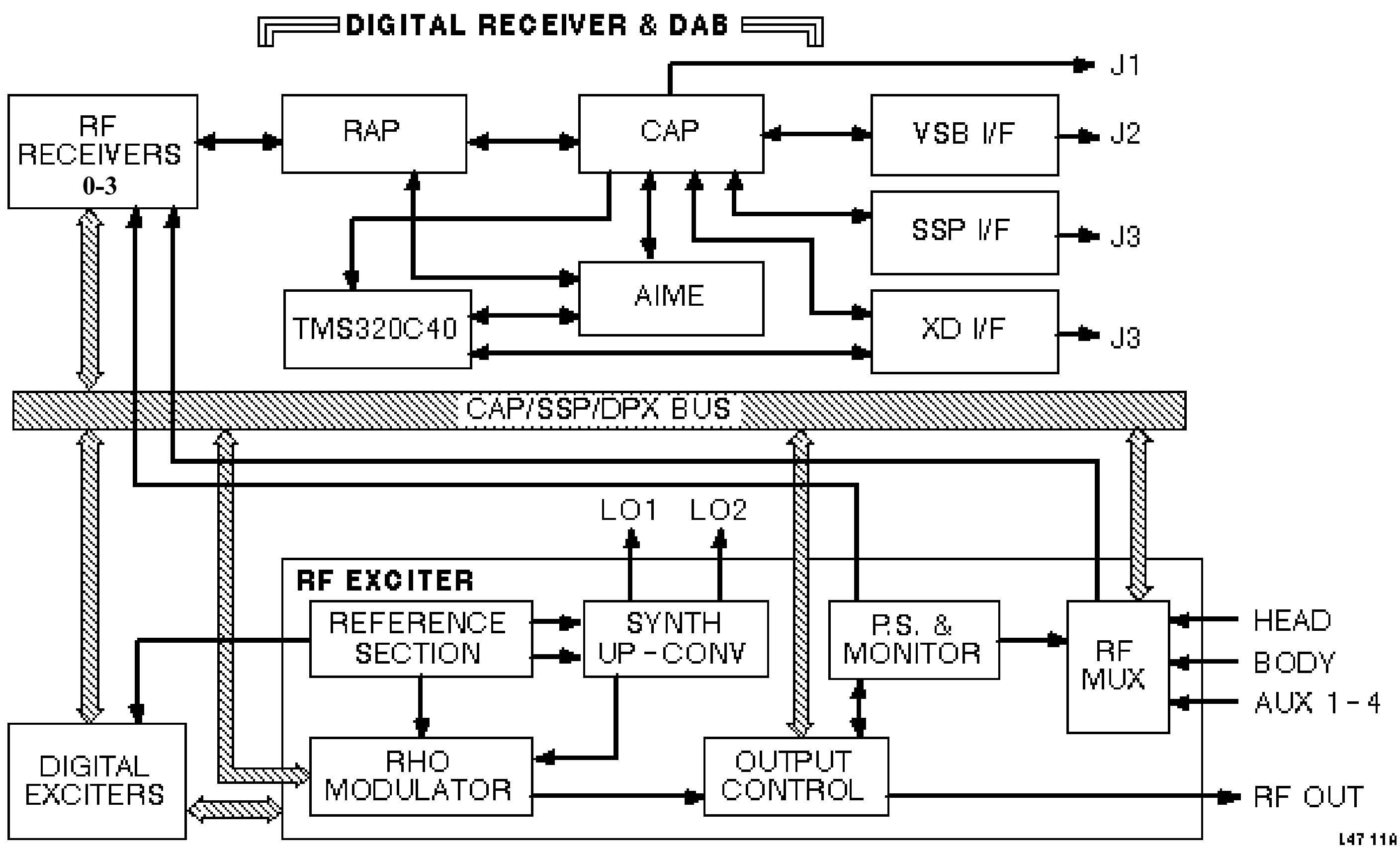
1 OVERVIEW

The Combined Exciter/Receiver DAB (CERD) is a combination of the Exciter, Receiver(s), and data acquisition board (DAB) in one triple-width VME module that occupies ISE chassis slots 13, 14, and 15. One-channel and four-channel versions are offered in the same physical package.

The CERD module consists of an extended 9U digital circuit board with a similar sized RF module mounted to it.

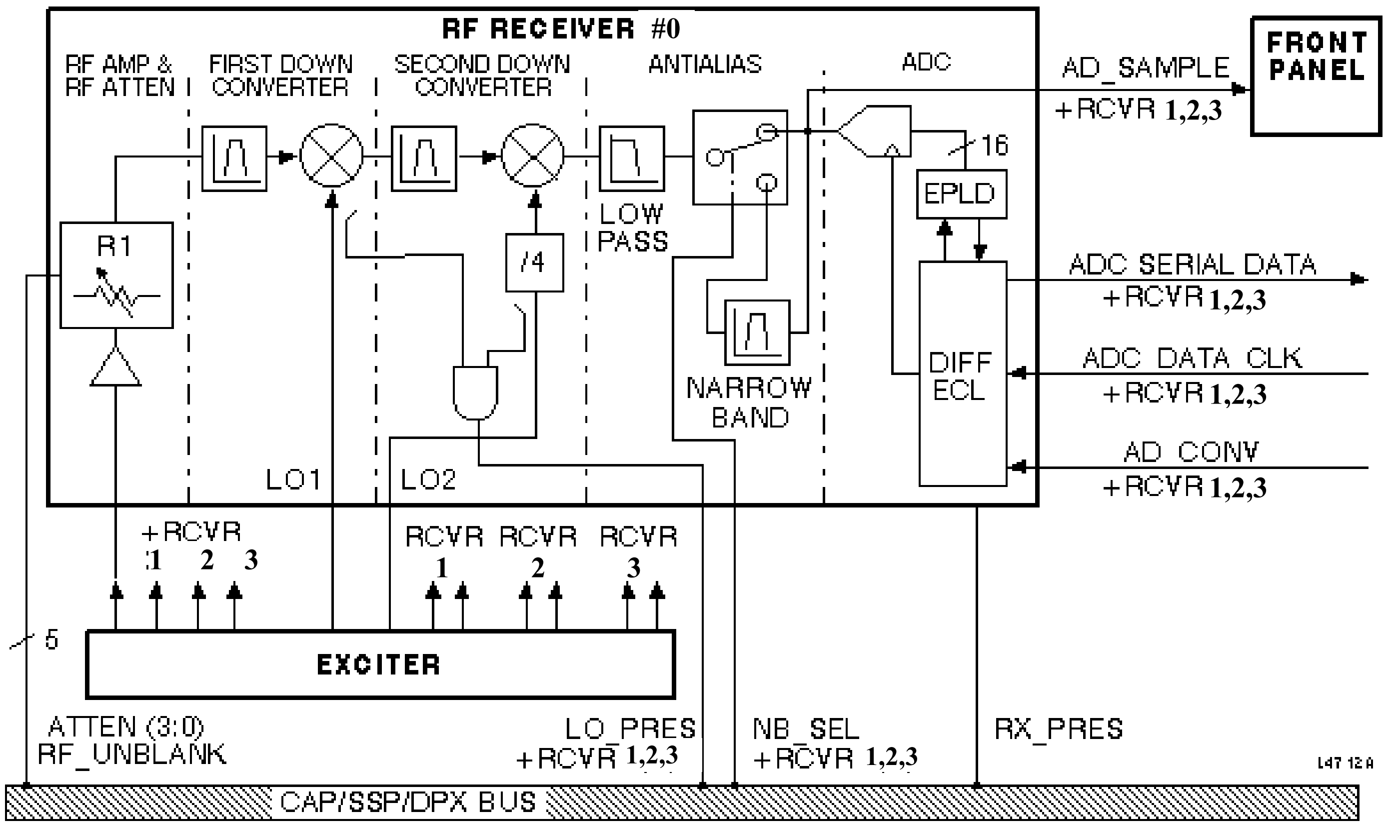
Upgrade from one to four channels is accomplished by plugging additional single-channel RF RX boards into the RF module.

The CERD provides a major advancement over the previous transceiver by including a high-speed, digital, frequency-domain processing chip set. Thus, the row FFT portion of reconstruction can be performed on the CERD. The CERD also has a mode that provides time-domain data to allow compatibility with current image reconstruction software.



CERD TOP LEVEL
ILLUSTRATION 1

2 RECEIVERS



RECEIVERS 1-4
ILLUSTRATION 2

2-1 RF Receiver Interface

In general, the firmware interface with the receiver functions may take one of the following forms:

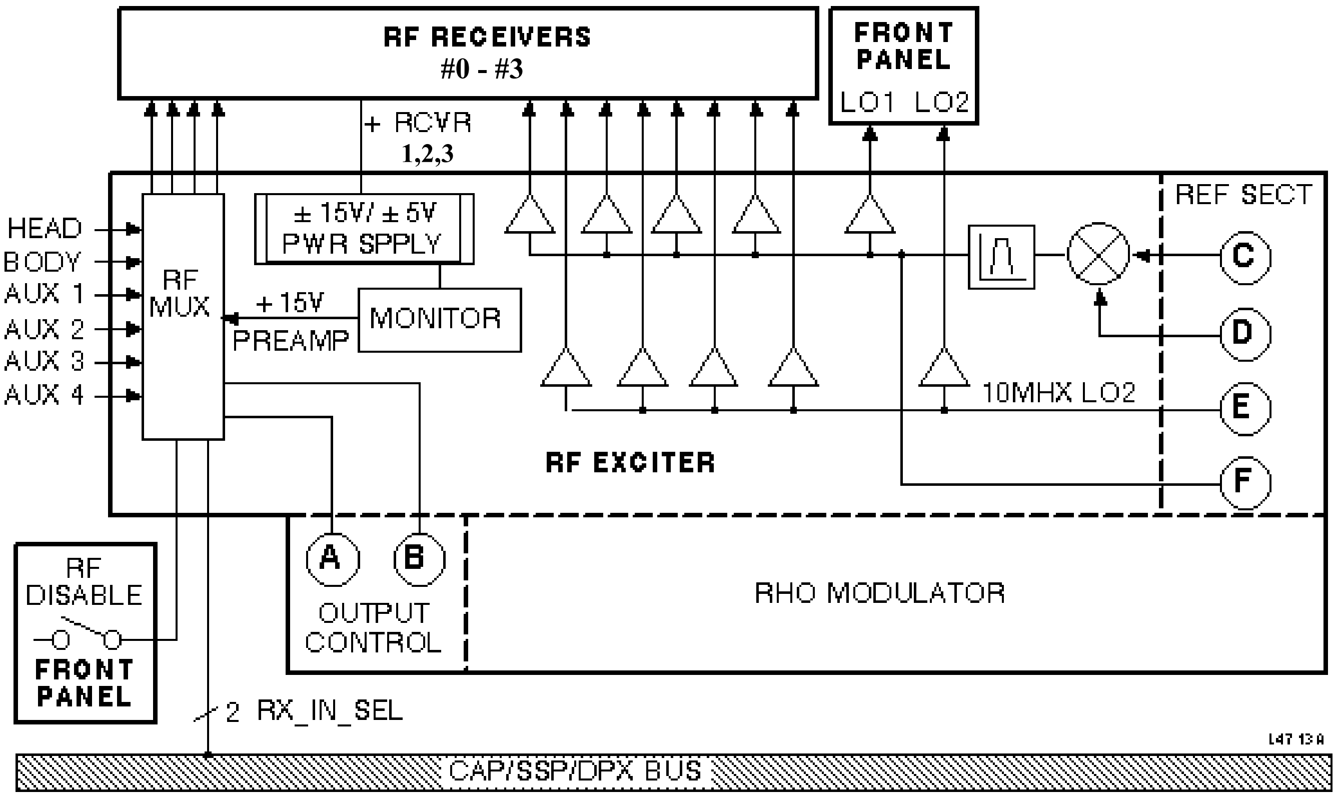
- CAP/SSP access
- CAP-only access
- ASC-only access

Synchronization is achieved for both the Exciter and Receiver registers by the one Synchronize command listed in the Exciter Memory Map. Setting asynchronous SSP operation is handled similarly.

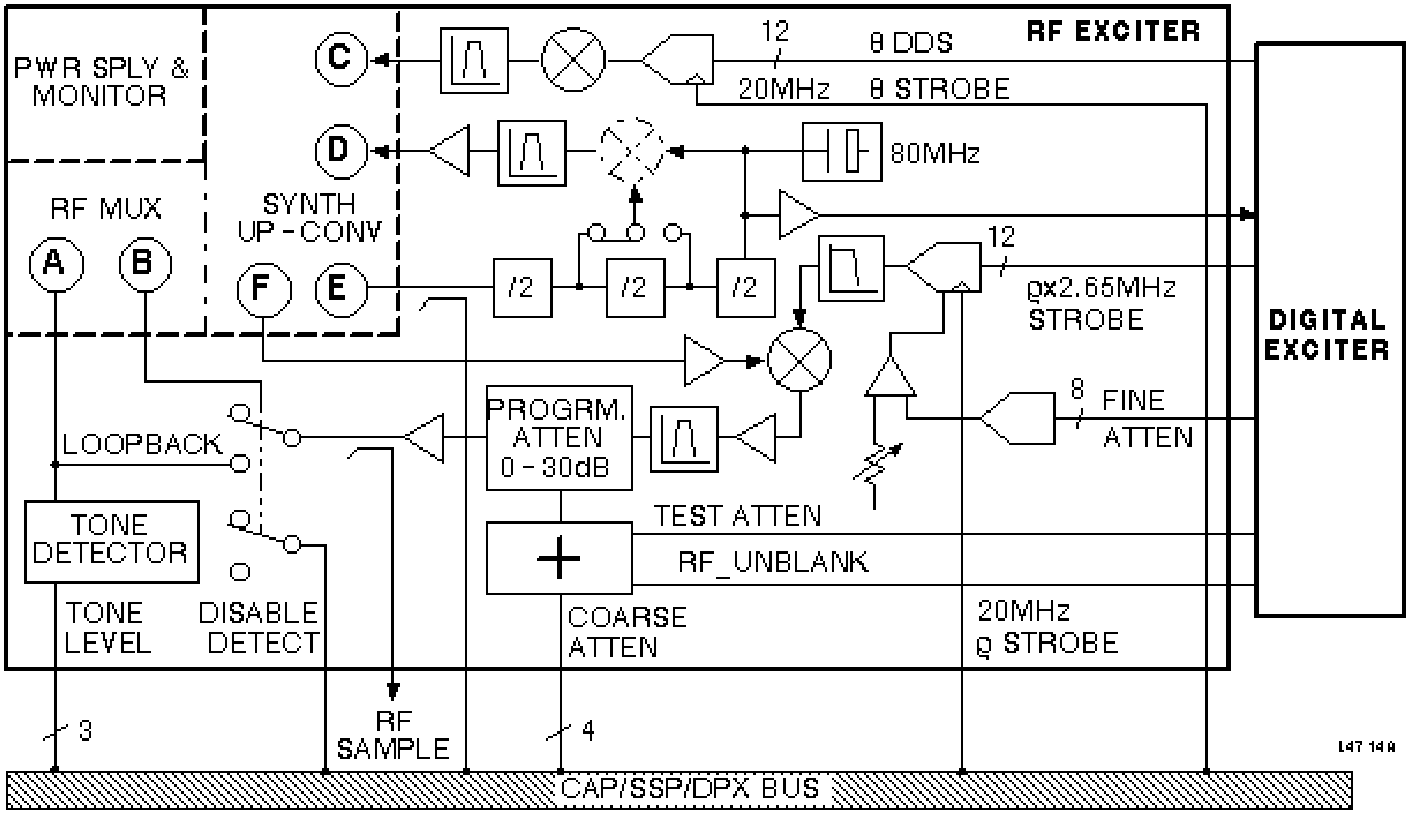
TABLE 1
RECEIVER #0 CONNECTOR

Pin #	Signal name	Signal Flow	Signal	Description
1	N/C			
2	N/C			
3	RX0_AD_CAL	Digital RF	Note 2	ADC calibrate
4	RX0_PRESENT	RF Digital	Note 2	Rx channel present
5	RX0_3dB	Digital RF	Note 2	R0 3dB
6	RX0_12dB	Digital RF	Note 2	R0 12dB
7	RX0_LO1_PRES	RF Digital	Note 2	LO1 present
8	LGND	Digital RF	GND	Logic GND through 1K pulldown
9	RX0_AD_CONV-N	Digital RF	Note 1	ADC convert strobe –
10	LGND	Digital RF	GND	Logic GND through 1K pulldown
11	RX0_DATA-N	RF Digital	Note 1,3	ADC serial data out –
12	LGND	Digital RF	GND	Logic GND through 1K pulldown
13	RX0_DATA_CK-N	Digital RF	Note 1	ADC serial data out clock –
14	N/C	Digital RF		
15	N/C	Digital RF		
16	RX0_UNBLNK	Digital RF	Note 2	RF module unblank
17	RX0_6dB	Digital RF	Note 2	R0 6dB
18	RX0_24dB	Digital RF	Note 2	R0 24dB
19	RX0_NB_SEL	Digital RF	Note 2	Select narrowband filter
20	LGND	Digital RF	GND	Logic GND through 1K pulldown
21	RX0_AD_CONV-P	Digital RF	Note 1	ADC convert strobe +
22	LGND	Digital RF	GND	Logic GND through 1K pulldown
23	RX0_DATA-P	RF Digital	Note 1,3	ADC serial data out +
24	LGND	Digital RF	GND	Logic GND through 1K pulldown
25	RX0_DATA_CK-P	Digital RF	Note 1	ADC serial data out clock +
Note 1: PECL (+5V supply differential ECL) with 100 parallel termination				
Note 2: TTL with input pulldown resistor				
Note 3: Output of differential receiver on digital board guaranteed to be logic 0 if RF receiver disconnected				
Receiver #1 Connector: Same as Receiver #0 except replace all instances of ”RX0” with ”RX1”				
Receiver #2 Connector: Same as Receiver #0 except replace all instances of ”RX0” with ”RX2”				
Receiver #3 Connector: Same as Receiver #0 except replace all instances of ”RX0” with ”RX3”				

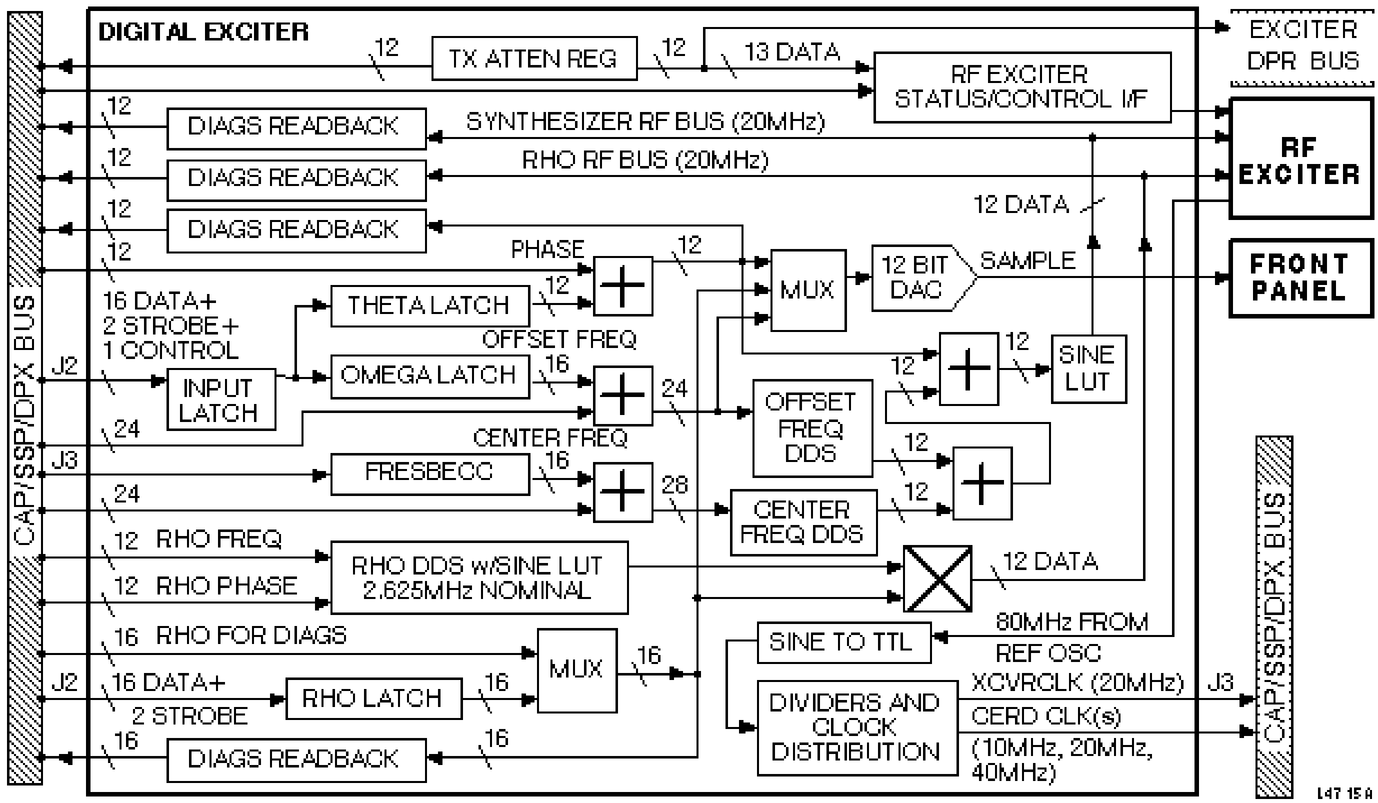
3 EXCITERS



RF EXC ITER 1
ILLUSTRATION 3



RF EXC ITER 2
ILLUSTRATION 4



DIGITAL EXCITER
ILLUSTRATION 5

TABLE 2
RF EXCITER BOARD CONNECTORS

Jumper #	Signal Name / Function	Connection Type	Signal Type
201	A/D Sample 1	BNC	Analog 5V (1K series resistor)
202	A/D Sample 2	BNC	Analog 5V (1K series resistor)
203	A/D Sample 3	BNC	Analog 5V (1K series resistor)
204	A/D Sample 4	BNC	Analog 5V (1K series resistor)
17	RF Sample Out	BNC	50 ohm out
18	LO1 Out	BNC	50 ohm out
19	LO2 Out (CERD Spectro Exciter)	BNC	50 ohm out
20	Analog power	25-pin sub-D, male	
21	RF Out to PA, Body, Head, AUX1, AUX2, AUX3, AUX4, +15VDC	RF coax D-shell, 8-pos	

TABLE 3
FRONT PANEL: DIGITAL BOARD CONNECTOR

Jumper #	Signal Name / Function	Connection Type	Signal Type
4	CAP Programmable	SMB	TTL out (1K series resistor)
5	RX Data In Window	SMB	TTL out (1K series resistor)
6	RF EX Unblank* (was TR Driver Unblank)	BNC	TTL out (1K series resistor)
7	RF RX Unblank*	BNC	TTL out (1K series resistor)
8	CAP Serial Port		TTL out (1K series resistor)
9	RHO/THETA/OMEGA Sample	BNC	Analog 5V (1K series resistor)
10	DABOUT 6 (= old DAB J10)	BNC	TTL out (1K series resistor)
11	DABOUT 7 (= old DAB J11)	BNC	TTL out (1K series resistor)
12	DABOUT 8 (= old DAB J12)	BNC	TTL out (1K series resistor)
The following signals currently available on the DAB are not available on the CERD front panel:			
DAB J5: DABOUT 1 (EFB EN*); available on TYME II Board			
DAB J6: DABOUT 2 (LOCKOUT*); available on TYME II Board			
DAB J7: DABOUT 3 (CINEFILT*); available on TYME II Board DAB J8: DABOUT 4 (SPECTRO UNBLANK); available on TYME II Board			
DAB J8: DABOUT 4 (SPECTRO UNBLANK); available on TYME II Board			
DAB J9: DABOUT 5 (SCOPE TRIG); available on IPG			

TABLE 4
FRONT PANEL: RF EXCITER BOARD POWER CONNECTOR

Power Supply Voltages	Pin-out* Test Points
+5VDC	1,2,14,15
-5.2VDC	5, 6, 8, 19
+15VDC	9, 10, 11, 12
-15VDC	22, 23
GND:	3,4,7,8,13,16,17,20,21,24,25
* Note: 25-pin sub-D connector	

4 RAP (RAW ACQUISITION PROCESSOR)

The function of the RAP is to collect data from the receivers, examine the data for overrange, offset correct the data, peak detect the data, test the data against a register-based overrange value, two's complement every other even and odd sample set when quadrature is turned on, queue the data for later use, and write the queued data to QIN memory.

When 1 or 4 CERD receivers are selected, the data from all four are transferred to the RAP (by the RAP) serially at 20 mHz. Because the RAP is generating the conversion and transfer clocks, collecting the CERD receiver data is synchronous.

The RAP performs offset correction to receive data. It does this by adding an offset value, from one of the four offset registers (BL0 through BL3), to the receive data. A value from a particular receiver, along with its offset register, is routed to an adder to obtain the corrected data. All four offset registers are used, along with data from the appropriate receivers during multichannel operation. Offset registers must be loaded with the appropriate values before acquisition begins.

The post offset corrected, peak absolute value from RCVR0 through RCVR3 is furnished via registers PK0 through PK3 respectively for readback by the ASC. The peak absolute value of LXD FIFO data will be on PK0. Because 1) both the data and its two's complement is available in the RAP, 2) implementing peak detection would double the number of peak registers required, and 3) peak detection would have required an additional adder to be implemented, the absolute value is used rather than ± peak values.

5 CAP (CONTROL ACQUISITION PROCESSOR)

The CAP is based on a TMS320C40 digital signal processor running at a clock frequency of 40 mHz. The CAP acts as the central processor for the CERD board, and interfaces with most of the CERD modules, including VME bus, VSB bus, SSP bus, ASC, AIME, RAP, and Exciter. The six communication ports that are used are:

- COM0 CAP to ASC COM 3
- COM1 CAP to LCA loader
- COM2 CAP to XD Bus LCA
- COM3 ASC COM 0 to CAP
- COM4 SSP PLD to CAP
- COM5 XD Bus LCA to CAP

6 VME INTERFACE

The VIC068A chip (Cypress Semiconductor) is used for the VME interface.

7 ADVANCED INTEGER MULTIPLIER ENGINE (AIME)

The AIME exists as a chip set using the Sharp LH9124 digital signal processor (DSP) and the Sharp LH9320 address generator (AG). The LH9124 is a high-performance, fixed-point DSP, optimized for DSP block-oriented algorithms and array processing. Real-time data processing is performed in either the time domain or the frequency domain. The LH9320 AG is a high-speed, memory-addressing device that creates address patterns for the LH9124 data. The AIME is controlled via the AIME Schedule Controller (ASC).

The ASC is based on a TMS320C40 digital signal processor running at a clock frequency of 40 mHz. The ASC acts as the control processor for the AIME, and interfaces with most of the CERD modules including the AIME, CAP, and RAP.

8 VSB INTERFACE

The VSB bus consists of a 32-bit multiplexed address/data bus, control signals for data transfer, and an arbitration bus.

The CERD VSB is a master-only interface. Only those features required for high performance data transfers are supported by it.

The CAP interfaces with the VSB bus through a bidirectional dual-port RAM (DPR). The RAM has two logical banks to allow the CAP to continue data transfers while the VSB bus is active.

The DPR size is 4K x 32 bits (1ch x 1frame/ch x 1k IQ pairs/frame x 2 lw/IQ pair x 2 = 4K lw). This assumes that buffering for all four channels exists between the AIME and the CAP.

A DMA controller is implemented in hardware to transfer data between CERD and BAM over the VSB bus. The CAP interfaces with the VSB DMA controller over a 16-bit data bus. Upon completion of the transfer(s), the VSB DMA controller asserts an interrupt to the CAP. If, for some reason, no VSB slave responds to the VSB DMA controller within 1.6 µs, an interrupt is generated to the CAP, indicating a transfer error. The transfer is terminated.

Except in *repeat data mode*, the DMA controller automatically increments the DPR address after each transfer. Thus, values are transferred to BAM in the order in which they appear in DPR. In repeat data mode, one value in DPR is written to many locations in BAM, providing a "fill" feature.

9 SSP INTERFACE

The CERD can receive only the SSP bus. For SSP diagnostics, the IPG drives the SSP bus.

The time base for time stamping is the internal CAP timer. The CAP normally receives only DAB packets, in order to minimize processor loading. In SSP monitor mode, all packets, independent of device code, need to be routed to the CAP. Resources normally devoted to processing scan data are available to handle SSP data in this mode.

CERD Exciter and Receiver packets still need to be routed to the hardware in this mode.

The main function of the SSP PLD is to multiplex SSP and CAP writes over the LSSP (Local SSP) bus. The SSP is a real-time bus that cannot be allowed to be held off by the CAP. The SSP PLD contains a state machine that provides alternating time slots for multiplexing the CAP/SSP data, and generating 100-ns write strobes. A CAP, or SSP write (to a valid address), causes a strobe to be generated sometime after the write. A flag is provided (STAT0), which is set after a CAP write to indicate that the buffered CAP data have not been sent over the LSSP bus. The CAP should not perform additional writes to devices on the LSSP bus until the CAP sees this flag reset. SSP writes to the CERD cause the state machine to momentarily reset if CTL4 is set, or no data was sent to CERD address FH (to enter synchronous SSP mode). If CTL4 is reset (default), and data have been written to CERD address FH, the state machine generates an SSP time slot every one millisecond, regardless of when the SSP write occurs. This SSP time slot is displaced several hundred nanoseconds from the synchronizing SSP write to allow variation in future SSP writes without affecting the actual register update time.

The SSP PLD decodes CAP read-addresses and provides the output-enable signals for devices on the LSSP bus, but the data from these devices do not flow through the PLD.

A strobe for an off chip *DAB_OUT* register is generated inside this PLD. A CAP comm port is connected to the PLD to send SSP data to the CAP when SSP monitor mode is selected, or when the CAP/DAB device code is selected via the SSP.

10 XD BUS INTERFACE

Table 6 shows the XD Bus modes that are supported, in the priority order listed. Only mode 4 is supported in Phase 1 models.

Only the LSW (first two bytes transferred over the ASC comm port) are pushed to the LXD FIFO in mode 3 operation. The transfer to FIFO is done after all four bytes have arrived over the comm port. The transfer rate for each burst of four bytes via comm port to the XD LCA is TBD versus the10 mb/s achieved in CAP to ASC (C40 to C40) communications.

TABLE 6
XD INTERFACE BUS MODES

Number	Name	Data Flow	Description	Priority
0	Receive & Filter	XD bus LXD (Local XD) FIFO (16-bit data)	Receive data from another device, e.g., Spectro Rx. Routes data to RAP for filtering by AIME.	3
1	Receive	XD bus CAP (16/32-bit data)	Receive data from another device, e.g., Spectro Rx. This matches operation of the current DAB.	2
2	Broadcast	CAP XD bus (16/32-bit data)	Broadcast data out, but the CERD supplies its own acknowledge. This allows the IPG to monitor the Rx data.	4

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 3, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Feb. 23, 1999	R. Kaufman	Changed Receiver numbering scheme to 0-3
2	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.